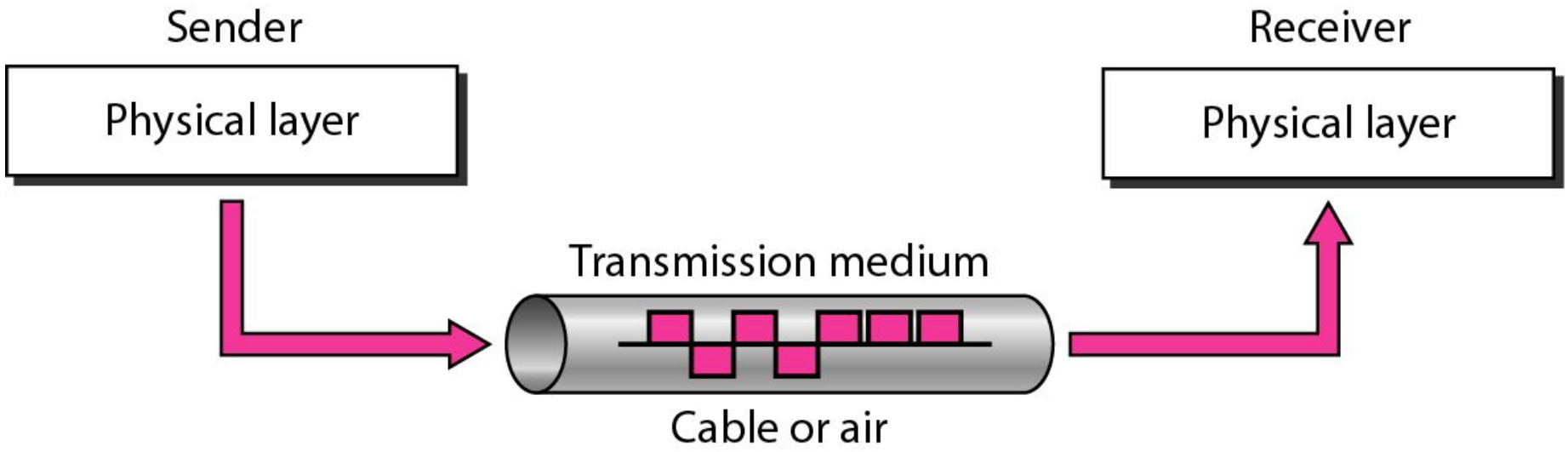
Data Communication
CSE 2211

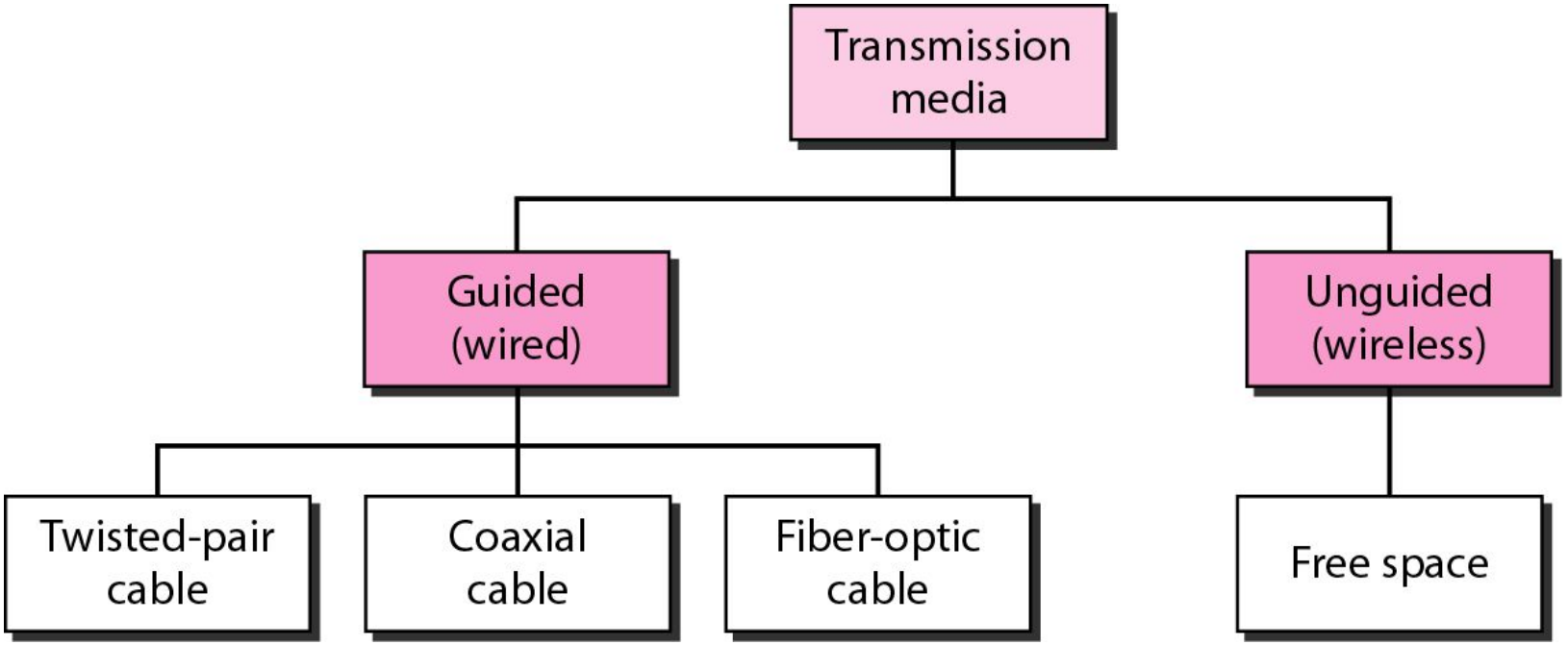
Chapter 7 : Transmission Media (Guided)

Transmission medium is the physical path between transmitter and receiver.

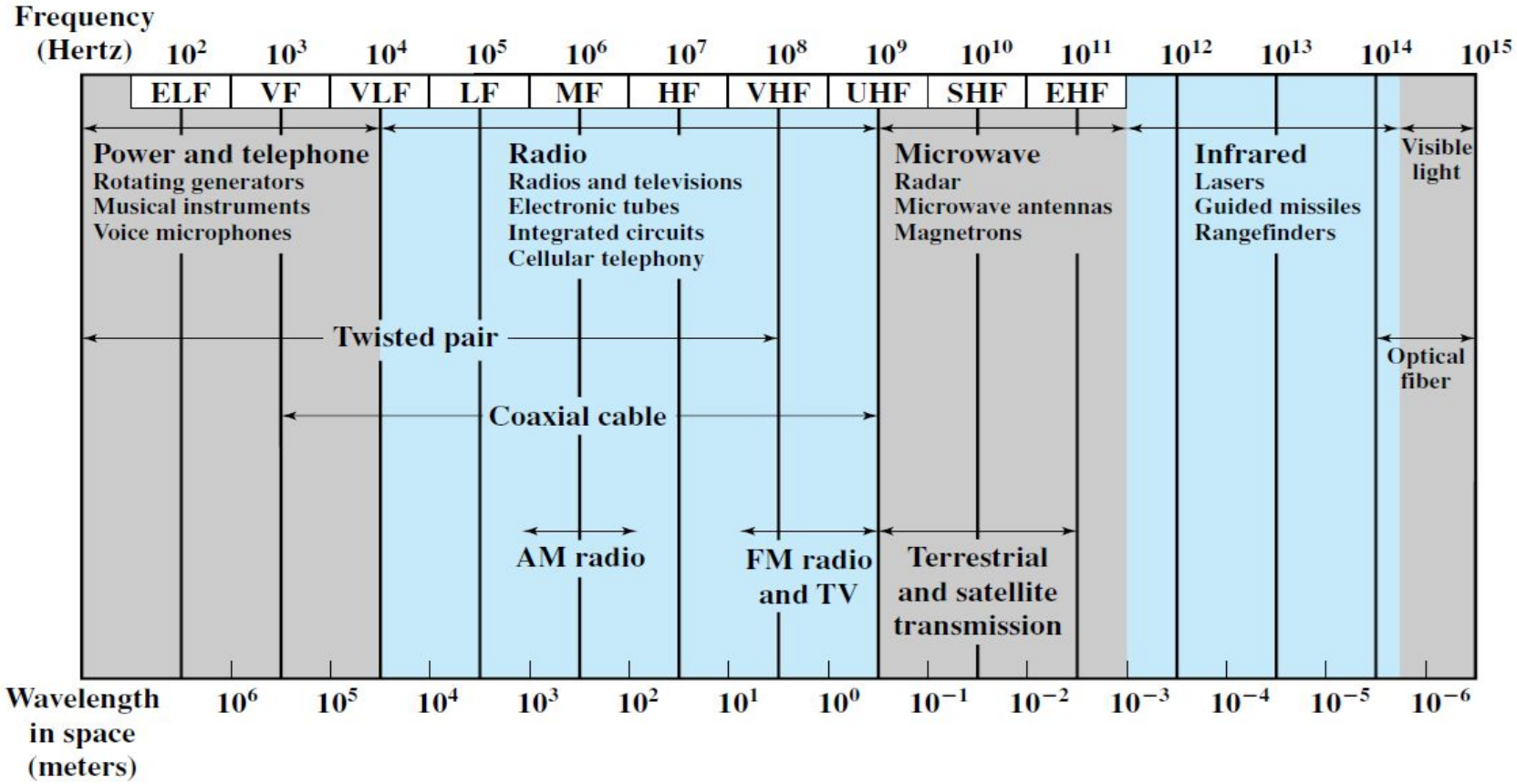
- Located below the physical layer and are directly controlled by the physical layer.



Classes of transmission media



Electromagnetic Spectrum



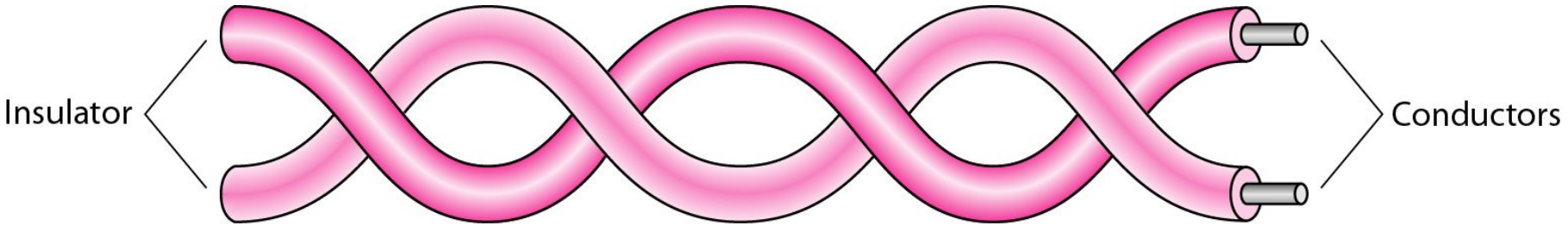
ELF = Extremely low frequency
VF = Voice frequency
VLF = Very low frequency
LF = Low frequency
MF = Medium frequency
HF = High frequency
VHF = Very high frequency
UHF = Ultra high frequency
SHF = Super high frequency
EHF = Extremely high frequency

7-1 GUIDED MEDIA

- *Waves are guided along a physical path*
- *Use a conductor such as a wire or a fiber optic cable to move the signal from sender to receiver*
- *Twisted-pair cables and coaxial cable: use metallic (copper) conductors that transport signals in the form of electric current*
- *Optical fiber : transport signals in the form of the light*

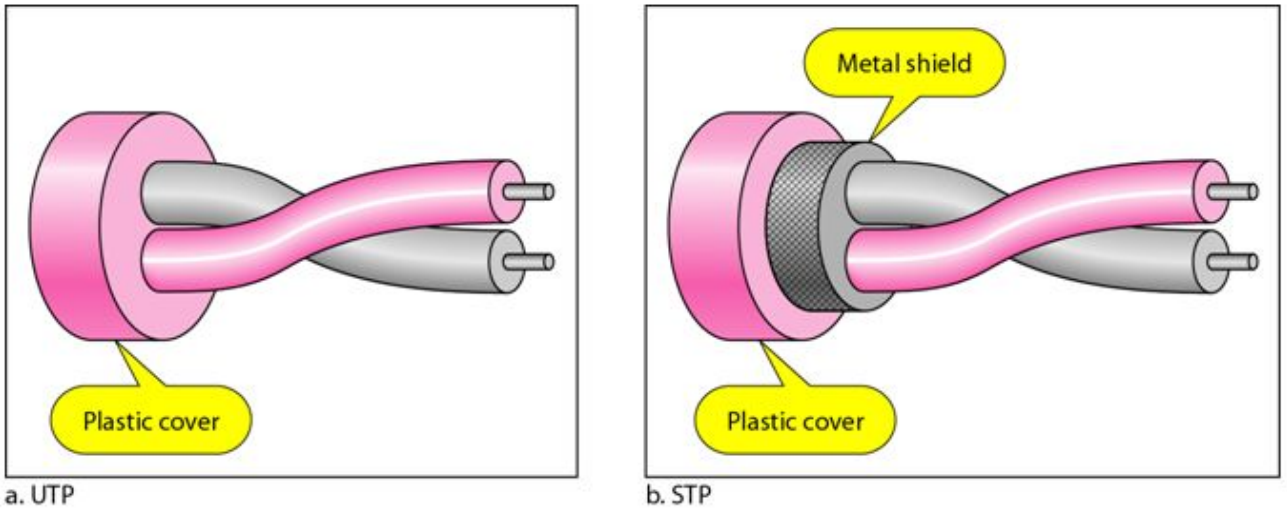
Twisted-pair cable

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of the wire used to carry signal and the other as a ground. The receiver uses the difference between the two.
- If the two wires are parallel, the effect of interference, noise and crosstalk is high.
- Twisting the pair of wire balance the effect of unwanted signal and reduce it.



Types of twisted pair

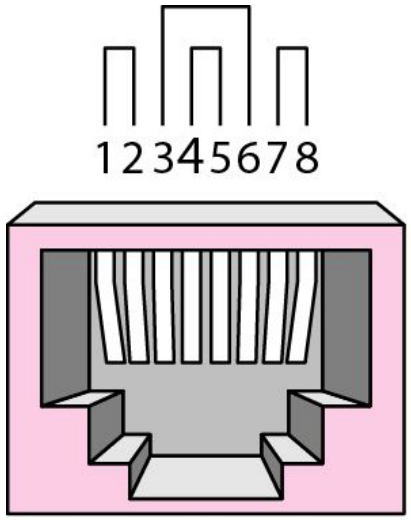
- *STP (shielded twisted pair)*
  the pair is wrapped with metallic foil to insulate the pair from electromagnetic interference
- *UTP (unshielded twisted pair)*
  each wire is insulated with plastic wrap, but the pair is encased in an outer covering



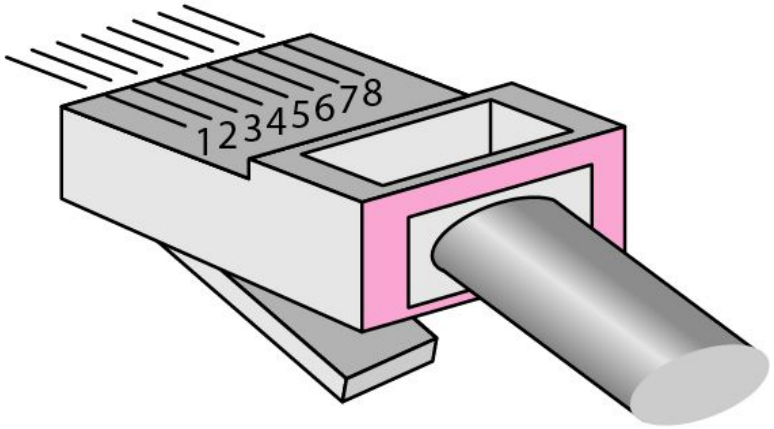
a. UTP

b. STP

Figure: *UTP connector*



RJ-45 Female



RJ-45 Male

Applications of Twisted-pair cable

Used in

1. Telephone lines to provide voice and data channels (local loop)
2. The DSL lines that are used by the telephone companies to provide high-data-rate connections
3. Local area networks, such as 10-base-T and 100base-T

Advantages and Disadvantages

- ***Twisted Pair Advantages***

Inexpensive and readily available
Flexible and light weight
Easy to work with and install

- ***Twisted Pair Disadvantages***

Susceptibility to interference and noise
Attenuation problem
    For analog, repeaters needed every 5-6km
    For digital, repeaters needed every 2-3km

Coaxial cable

- Coax cable carries signals of higher frequency ranges than those in Twisted pair cable.
- Instead of having two wires, coax has a central core conductor of solid which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two.
- The outer conductor serves both as a shield against noise and as second conductor, which complete the circuit.

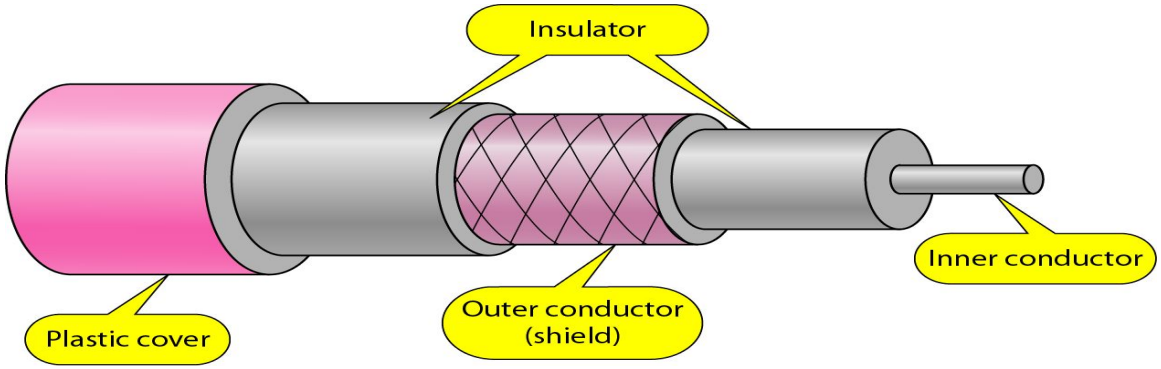
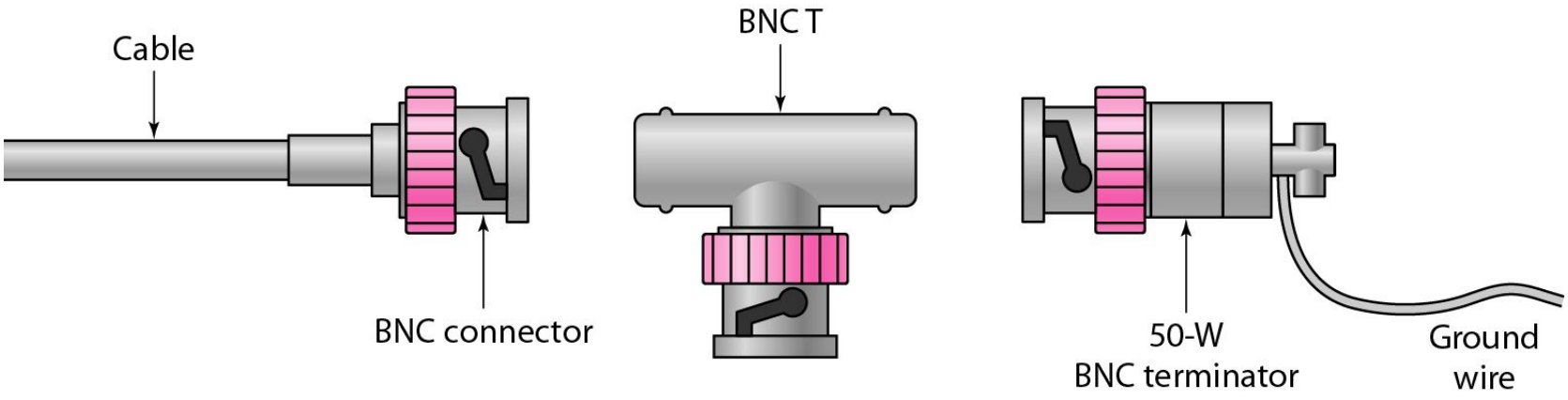


Figure: *BNC connectors*



Applications of Coaxial cable

1. **Analog telephone network where a single cable could carry 10,000 voice signals. Later it was used in Digital telephone networks where cable can carry 600Mbps**

2. **Cable TV network: In the traditional cable TV network, the entire network used RG-59 coaxial cable.**

3. **Traditional Ethernet LANs.**

   10-base-2 or "Thin Ethernet", uses RG-58 coax cable to transmit data at 10 Mbps with a range of 185m.

   10-base-5 or "Thick Ethernet", uses RG-11 to transmit 10 Mbps with rang of 500 m

Advantages

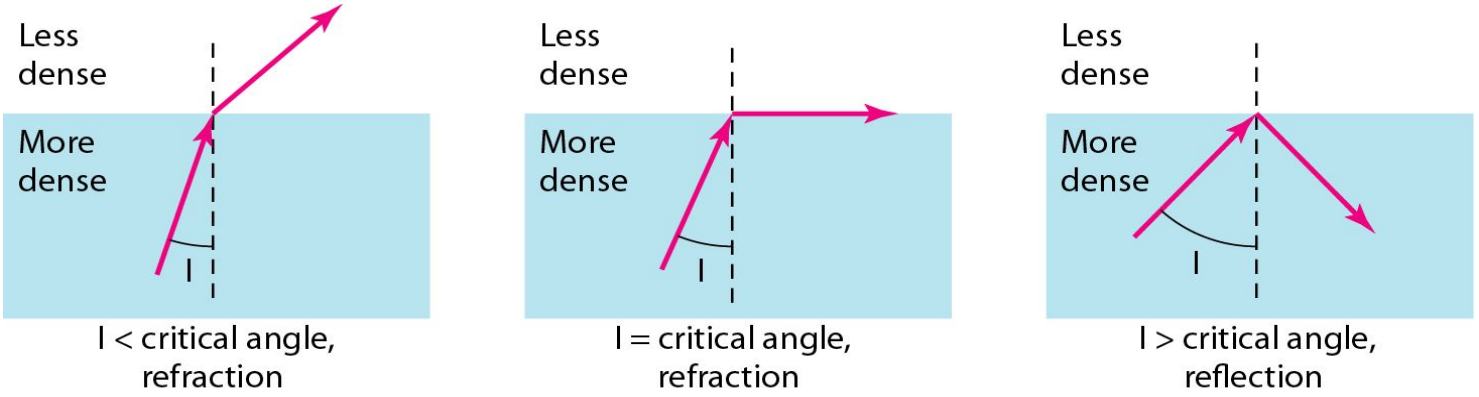
- *Higher bandwidth*
  - *400 to 600Mhz*
  - *up to 10,800 voice conversations*
- *Can be tapped easily (pros and cons)*
- *Much less susceptible to interference than twisted pair*

Disadvantages

- *More expensive to install compare to twisted pair cable.*
- *The thicker the cable, the more difficult to work with.*

Fiber-optic cable

- A fiber-optic cable is made of glass or plastic and transmit signals in the form of light.
- Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance enters another substance of different density , the ray change direction as shown:



- If the **angle of incidence** *I* is less than the **critical angle,** the ray **refracts** and moves closer to the surface.
- If the angle of incidence is equal to the critical angle, the light bends along the interface.
- If the angle is greater than the critical angle, the ray **reflects** (makes a turn) and travels again in the denser substance.

Figure 7.11 *Optical fiber*

- Optical fibers use reflection to guide light through a channel.
- A glass or plastic **core** is surrounded by a **cladding** of less dense glass or plastic.
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

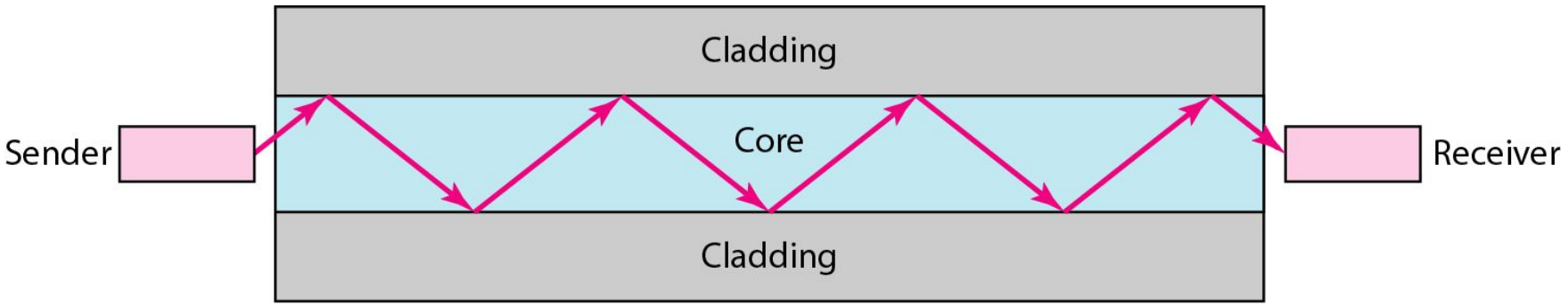
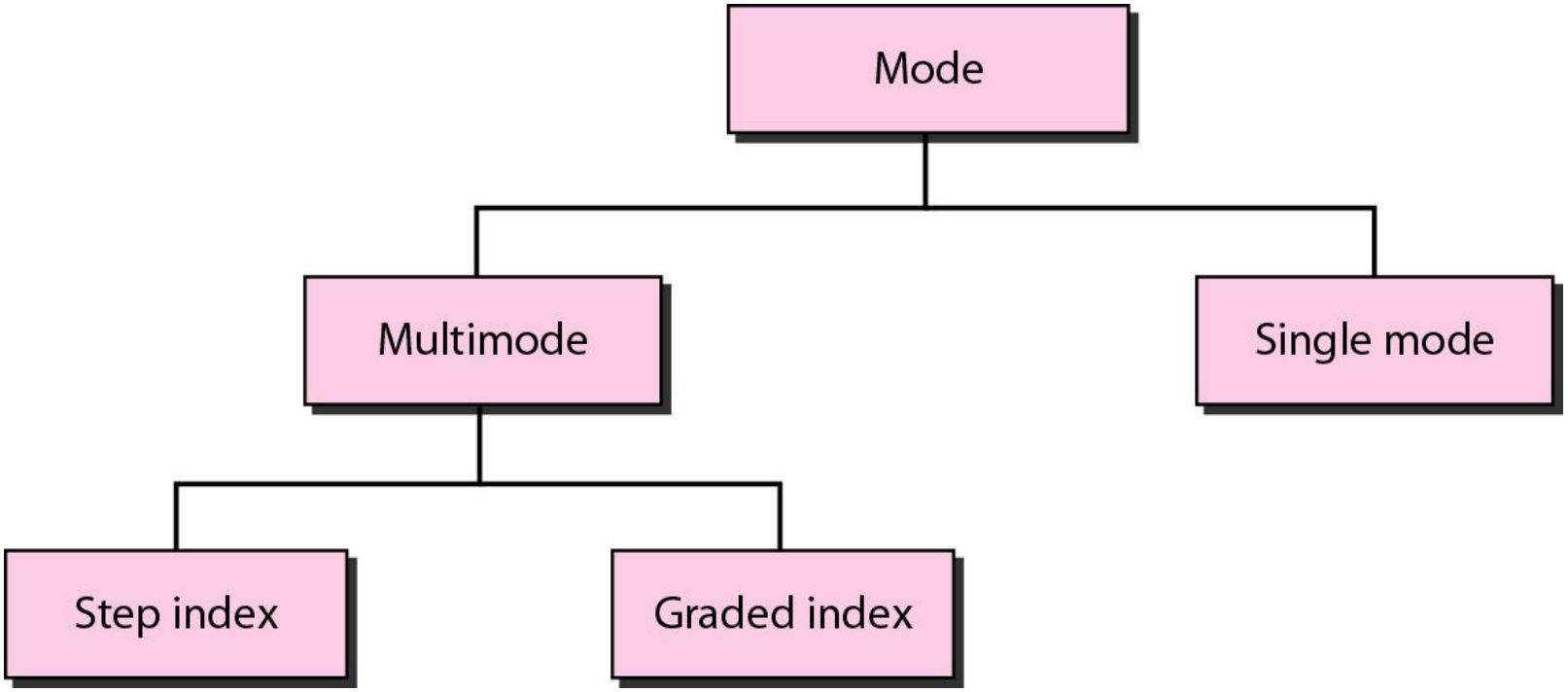


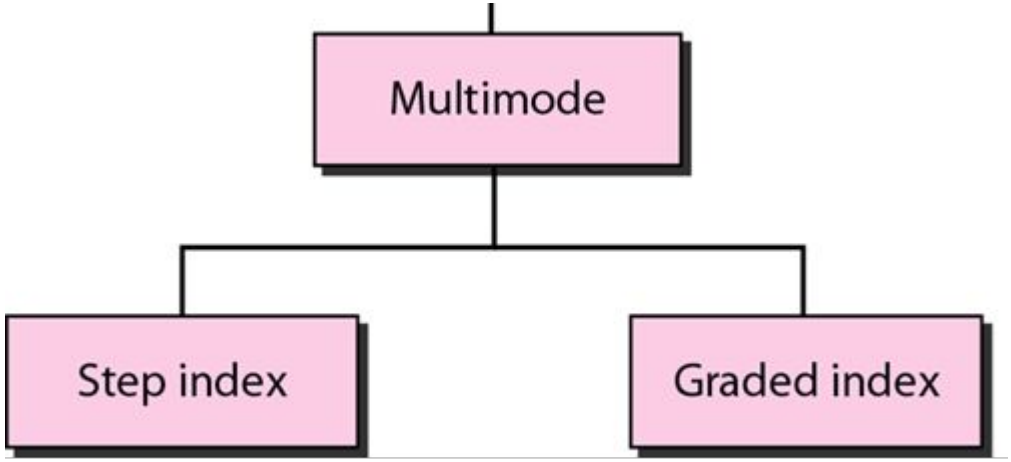
Figure 7.12 *Propagation modes*



Multimode

- *Multimode is so named because multiple beams from a light source move through the core in different paths.*

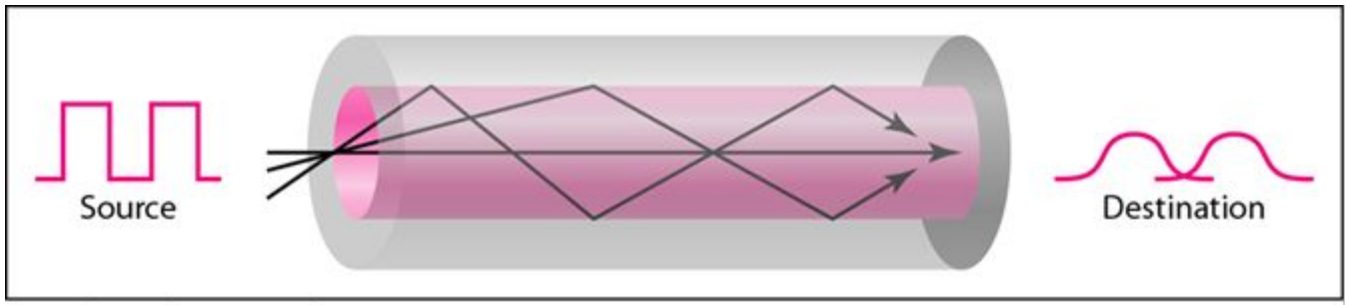
- *How these beams move within the cable depends on the structure of the core.*



Multimode Step-index fibre

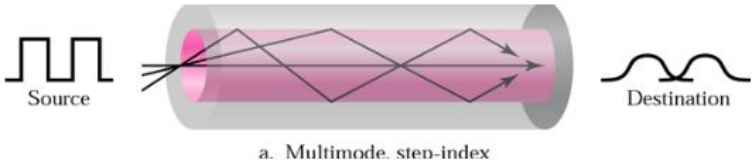
- *In multimode step-index fiber, the density of the core remains constant from the centre to the edges.*

- *A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.*



a. Multimode, step index

Multimode step index

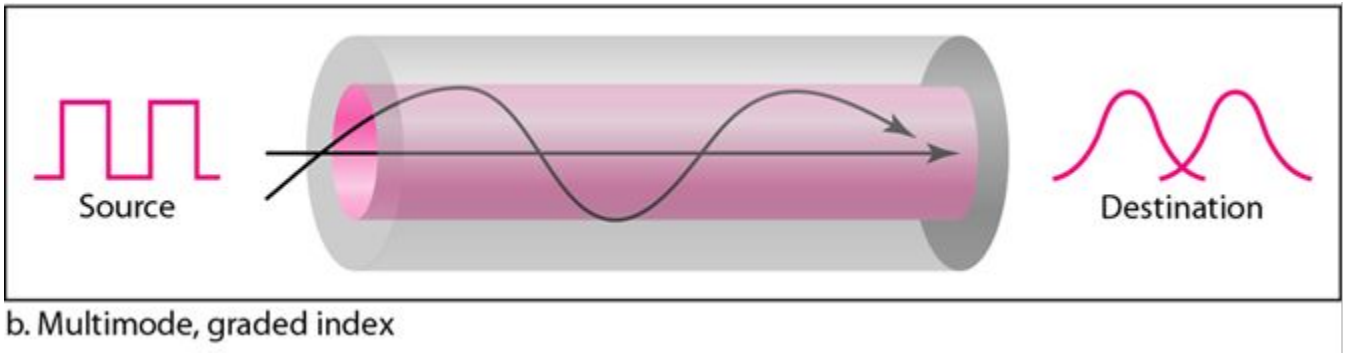


a. Multimode, step-index

- *Some beams in the middle travel in straight lines through the core and reach the destination without reflecting and refracting.*
- *Some beams strike the interface of the core and cladding at an angle smaller than the critical angle; these beams penetrate the cladding and are lost.*
- *Still other hit the edge of the core at angles greater than the critical angle and reflect back into the core and off the other side, bouncing back and forth down the channel until they reach the destination.*

Multimode Graded-index fibre

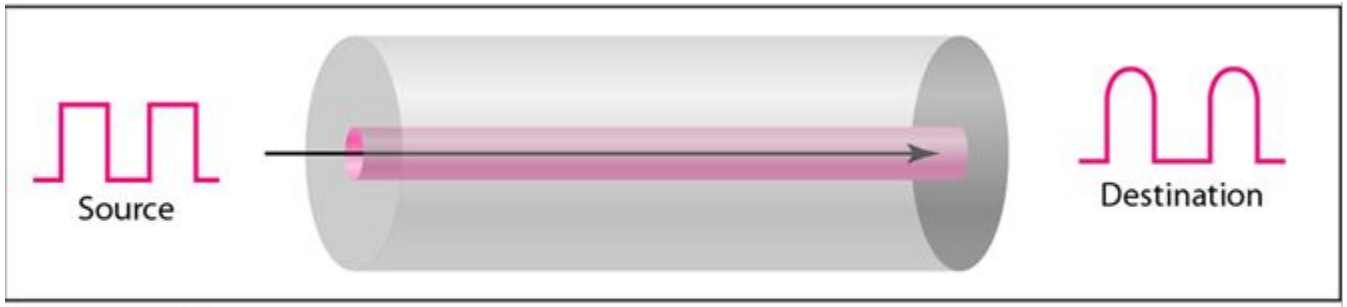
- *A graded-index fiber is one with varying densities.*
- *Density is highest at the centre of the core and decreases gradually to its lowest at the edge.*
- *The impact of this variable density on the propagation of light beams is shown below:*



b. Multimode, graded index

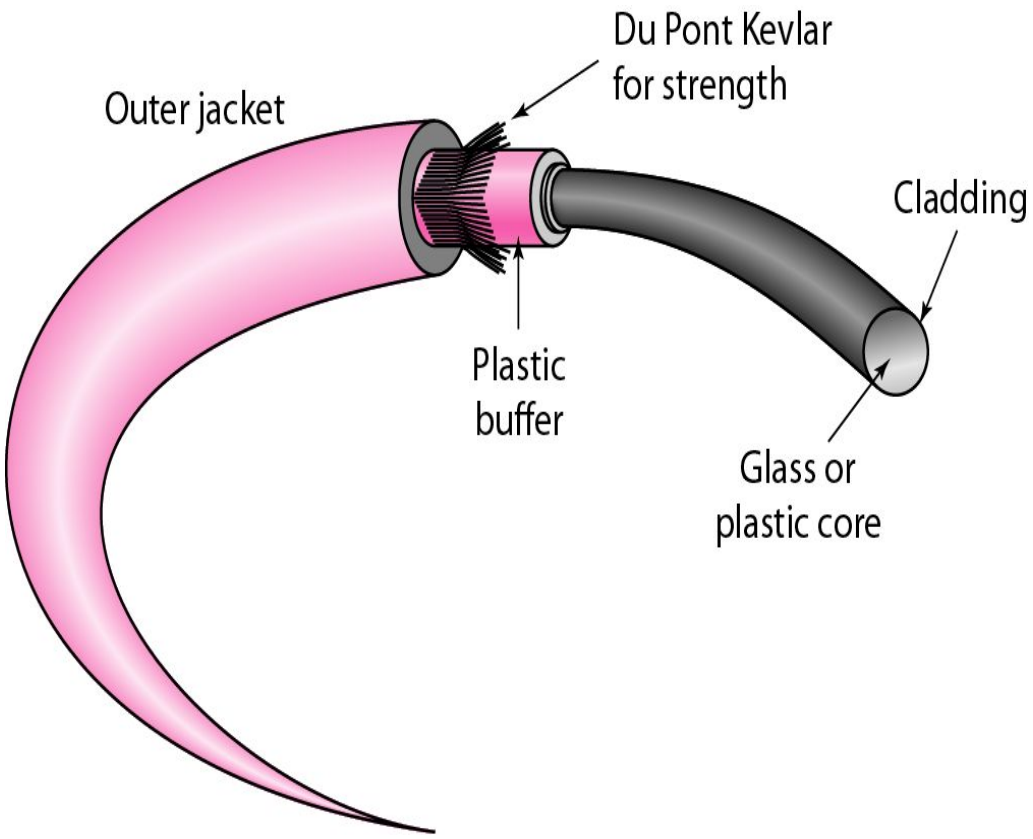
Single Mode

- *Single mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles and close to the horizontal.*
- *The single mode fiber itself is manufactured with a smaller diameter than that of multimode fibers, with substantially lower density.*
- *The decrease of density results in a critical angle that is close enough to 90 degrees to make the propagation of beams almost horizontal.*



c. Single mode

Figure: *Fiber construction*



The outer jacket is made of either PVC or Teflon.

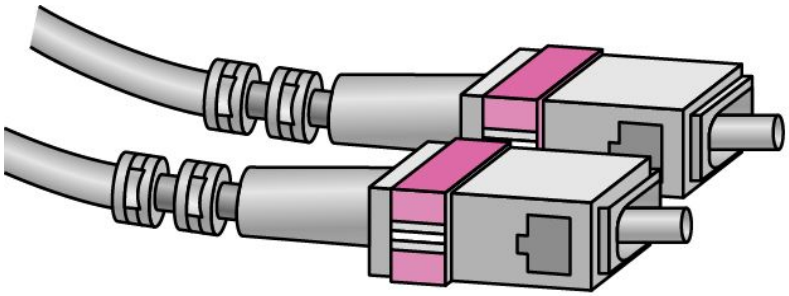
Inside the jacket are Kevlar strands to strengthen the cable. Kevlar is a strong material used in the fabrication of bulletproof vests.

Below the Kevlar is another plastic coating to cushion the fiber.

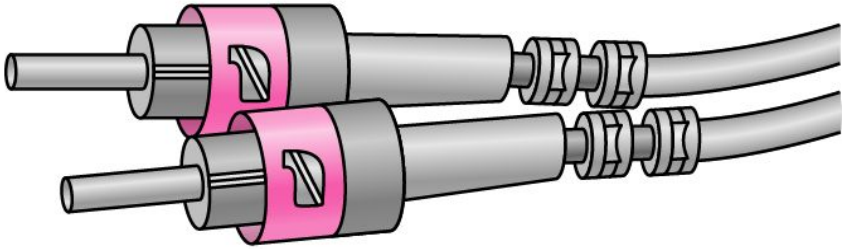
The fiber is at the center of the cable, and it consists of cladding and core.

Figure: *Fiber-optic cable connectors*

Subscriber channel (SC) connector is used for cable TV

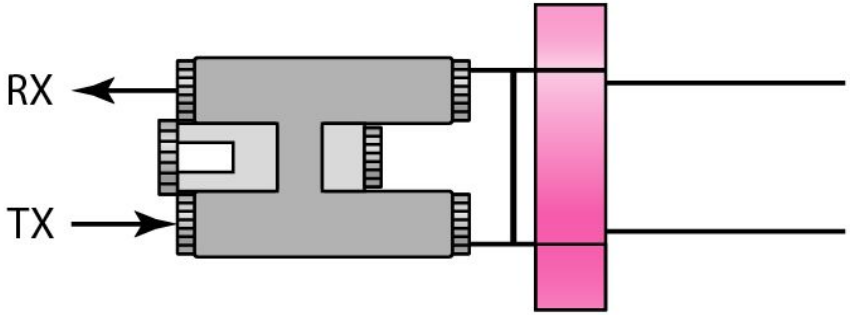


SC connector



ST connector

Straight-tip (ST) connector is used for connecting cable to networking devices.



MT-RJ connector

MT-RJ is a connector that is the same size as RJ45

Applications of Fiber-optic cable

Used in

1. **Cable TV network: hybrid network use a combination of optical fiber and coax cable. Optical provides the backbone while coaxial cable provide the connation to the user.**

2. **Local area networks such as 100base-FX(fast Ethernet) and 1000base-XLANs.**

3. **Backbone networks because its wide bandwidth**

Advantages of fiber-optical

1. **Higher Bandwidth**

2. **Less signal attenuation**

3. **Immunity to electromagnetic interference (noise)**

4. **Resistance to corrosive materials. Glass is more resistance to corrosive material**

5. **Light weight. Fiber cables are much lighter than copper cables**

6. **Greater immunity to tapping: copper cables create antenna effects that can easily tapped**

Disadvantages of fiber-optical

1. **Installation and maintenance.** It's a new technology. Its installation and maintenance require expertise that is not yet available everywhere.

2. **Unidirectional light propagation.** If we need bidirectional , two fibers are needed.

3. **Cost.** The cable and the interfaces are more expensive than those of other guided media. If the demand of BW is not high , often use of optical fiber can not be justified